

Jisung Kim

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RESEARCH INTERESTS

- **Computational Fluid Dynamics (CFD)**
- **Turbulent Flow Modeling**
- **Biomimetics**

Computational fluid dynamics (CFD), turbulent flow modeling, and biomimetic approaches for high-Reynolds number and transitional flows in aerospace applications, with emphasis on atmospheric entry and boundary layer phenomena.

EDUCATION

Kyungpook National University

Department of Astronomy and Atmospheric Sciences
GPA: 3.34/4.3

Expected Graduation Date: February 2026

Gyeongsang National University

Department of Geology

03/2020 – 12/2023

(Note: Military Service, Republic of Korea Army, 04/2021-10/2022)

INTERSHIPS

Undergraduate Researcher

Atmospheric Boundary Layer Laboratory

03/2024 – Present

- **Turbulence Structure Analysis:** Conducted comprehensive studies on turbulent flow characteristics in atmospheric boundary layers, analyzing Reynolds stress tensor components and energy dissipation mechanisms applicable to aerospace boundary layer problems.
- **Flux-Variance Relationship Investigation:** Quantified dissimilarity patterns in scalar transport (temperature, humidity, CO_2) under varying atmospheric stability conditions, providing insights for heat transfer modeling.
- **Advanced Data processing:** Implemented real-time quality control algorithms for wind observation tower data using Python and MATLAB, processing high-frequency (10Hz) turbulence measurements.

PROJECT EXPERIENCES

• Structural and Operational Analysis of American Airlines Flight 587 Aviation Accident

-Introduction of Aerospace Engineering

Fall 2024

- Investigated vertical stabilizer failure due to excessive rudder deflection and composite material stress.
- Identified issues in wake turbulence response and training programs.
- Proposed safety recommendations based on the Cooper-Harper pilot workload scale.

• Aerodynamic and Structural Analysis of A-10 Thunderbolt II

- Introduction of Aerospace Engineering

Fall 2024

- Analyzed the aerodynamic properties of NACA 6716 airfoil (high-lift coefficient, high L/D ratio) relevant to CAS performance.
- Evaluated structural materials (Al 7075-T6, Ti alloy) and honeycomb construction for redundancy and survivability.

- Assessed engine performance (TF34-GE-100) under extreme conditions, including post-damage operability.

• Correlation between wind power generation and meteorological factors

-Micrometeorology

Fall 2024

- Analyzed wind power output and meteorological data to optimize renewable energy predictions in South Korea's top wind energy regions.
- Built a Python GIS tool to align wind turbines with AWS stations, improving meteorological data accuracy.
- Conducted time-series analysis using Python to visualize seasonal wind power trends, identifying winter peaks driven by micro-meteorological factors to inform turbine operation strategies.

• Computational Modeling of Atmospheric Optical Properties

-Atmospheric Radiation

Fall 2024

- Modified Fortran models to compute optical depth and transmittance, visualizing results with Python.
- Simulated spectral line broadening and water vapor effects using HITRAN, refining model predictions.
- Simulated infrared radiative transfer in the atmosphere by calculating key optical properties such as absorption, optical depth, and transmittance.

• Comparative Analysis of Nitrogen Dioxide and Ozone Trends Based on Traffic Volume and Solar Radiation

-Atmospheric Pollution

Fall 2024

- Built Fortran-based numerical models with implicit methods to compute concentration changes of ozone (O_3) and species like NO and NO_2 at 0.01-second intervals.
- Utilized Python to visualize temporal trends, highlighting ozone increase from 4.92778×10^{18} to $4.92779 \times 10^{18} \text{ molecules/cm}^3$ and key photochemical reactions.

• Ozone Photochemistry Data Analysis

-Atmospheric Chemistry

Spring 2024

- Processed atmospheric chemistry datasets using Fortran and Python to model and visualize ozone photochemistry dynamics.
- Built Fortran-based numerical models with implicit methods to compute concentration changes of ozone (O_3) and species like NO and NO_2 at 0.01-second intervals.

SKILLS

Programming Languages: Fortran, Python, Linux (Shell scripting), MATLAB

Others

Languages

- Korean (Native), English (Intermediate), German (Basic)

English Proficiency

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